

1 history

2 bandwidth

3 archive weights

4 proposal

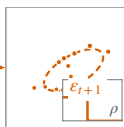
5 one draw

6 weight

(a) textbook ABC-PMC, schematic: one update per generation shown: generation $t \rightarrow t+1$, after 135 simulations



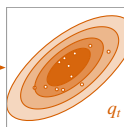
the current population only; earlier generations are discarded



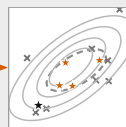
next threshold: a quantile of the population's discrepancies; accept or reject



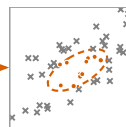
weights carried over from the last generation; covariance from them



one mixture proposal over the whole population



propose, simulate, accept or reject, one candidate at a time, until N are kept

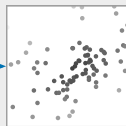


the kept are weighted against that one proposal, after acceptance

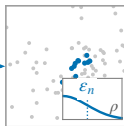
P_{t+1} : the N kept, weighted; the rejected are discarded

barrier: q_{t+1} needs the complete P_{t+1}

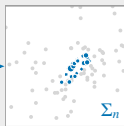
(b) generation-free (ours): one update per arrival shown: the n th arrival after 84 evaluations, $w^* = 1$, no bandwidth stored



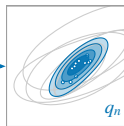
every particle ever evaluated, with its stored bandwidth and weight



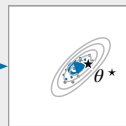
the k best form the archive; a smooth kernel of width ϵ_n , which only ever tightens



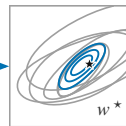
archive weights: stored weight times kernel weight; covariance from them



one mixture proposal over the archive, rebuilt at every call



one candidate: a parent drawn by weight, then perturbed



weighted against the current proposal and the S stored proposals, before it is simulated

θ^* to a free worker; ρ^* arrives later, among other workers' results: append $(\theta^*, \rho^*, \epsilon_n, w^*)$